Project 1 Data Card

Apple Financial Dataset

Authors	Last updated	Created
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2. Dataset at a glance
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Card ID

Primary Dataset	*Apple Financial Dataset (1980-2026)*
Description	*All-in-One Apple Stock Dataset: Prices, Financials, and ML Features (1980-2026)*
Supplemental Dataset 1	*Apple Stock (AAPL): Historical Financial News Data*
Description	*A comprehensive dataset of financial news on Apple Inc. (AAPL) from 2016 to 2024*
Supplemental Dataset 2	*Apple iPhone Product & Revenue Dataset (2007–2025)*
Description	*A yearly dataset capturing iPhone product releases, estimated sales, pricing, and revenue contribution from 2007 to 2025*
Supplemental Dataset 3	*Global iPhone & Smartphone Market Dataset (2011–2023)*
Description	*A global smartphone market dataset covering iPhone sales, users, and iOS vs. Android market share trends from 2011 to 2023*

Add more rows to the above table for each additional supplemental dataset.

Datasets at a glance

Primary dataset

The primary dataset contains 11,879 examples representing daily historical Apple stock records in the period from 12/12/1980 to 01/27/2026.

Fields (if applicable)

Field name	Data Type	Description
date	Date	Trading day timestamp; ranges 12/12/1980–01/27/2026. Example: 2024-11-22
close_price	Float	Closing stock price for the quarter; primary indicator of Apple's market value over time. Example: 0.116784.
open_price	Float	Opening stock price for the quarter; used to measure price movement relative to the closing price. Example: 0.098389.
total_volume	Integer	Number of shares traded; ranges 1M → 200M+ depending on year. Example: 42,300,000
avg_daily_return	Float	Average percentage change in price from one trading day to the next within the quarter; measures growth and momentum. Example: 0.015527
return_volatility	Float	Standard deviation of daily returns within the quarter; quantifies risk and the intensity of price fluctuations. Example: 0.049939

Supplemental dataset 1

The supplemental dataset contains news articles related to Apple Inc. (AAPL), representing financial and market-related coverage during the period from 01/01/2016 to 12/31/2024.

Fields (if applicable)

Field name	Data Type	Description
Date	Date	Publication date of the news article Example: 2024-11-26T00:00:00+00:00
Headline	String	Title of the news article Example: Could Investing $1,000 in Apple Make You a Millionaire?
Source	String	News publisher or media outlet

		Example: https://www.fool.com/investing/2024/11/26/could-investing-1000-apple-make-you-millionaire/
Summary	String	Short summary or excerpt of the article Example: A Complete Guide to College Savings Accounts in 2024
Sentiment	Float	Sentiment score associated with the article Example: 0

Supplemental dataset 2

The supplemental dataset contains annual observations representing the iPhone's product releases, sales performance, and revenue impact on Apple during the period from 01/01/2007 to 12/31/2025.

Fields (if applicable)

Field name	Data Type	Description
Year	Integer	Calendar year of observation Example: 2007
iPhone_Model_Released	String	Flagship iPhone model(s) released in the given year Example: iPhone 3G
Units_Sold_Millions	Float	Estimated number of iPhones sold in the year (in millions) Example: 1.99
Average_Selling_Price_USD	Float	Estimated average selling price of iPhones (USD) Example: 499
Annual_iPhone_Revenue_Billions	Float	Estimated total iPhone revenue for the year (USD billions) Example: 0.95

Supplemental dataset 3

The supplemental dataset contains annual observations representing global iPhone sales, user adoption, and smartphone market share trends during the period from 01/01/2011 to 12/31/2023.

Fields (if applicable)

Field name	Data Type	Description

Year	Integer	Calendar year of observation Example: 2023
Global_iPhone_Users_Millions	Float	Estimated number of iPhone users worldwide (in millions) Example: 1460000000
iPhone_Sales_Millions	Float	Annual global iPhone unit sales Example: 234600000
iOS_Market_Share_Percent	Float	Global smartphone market share of iOS (%) Example: 58.10
Android_Market_Share_Percent	Float	Global smartphone market share of Android (%) Example: 41.46

Additional notes

Provide any additional notes on the data here (e.g. changes in data collection and labeling practices).

Usage information

Use(s)

Primary Data
Usage for the dataset includes time-series forecasting, or analyzing the correlation between financial reports and stock price movements and for educational and research purposes. Discouraged use cases are using data for real-time trading and investment decisions, official financial statements, and high-stakes predictive modeling.

1st Supplementary Data
Usage for the dataset includes analyzing financial news sentiment related to Apple Inc., studying the relationship between media coverage and stock price movements, and developing NLP-based sentiment analysis models for educational and research purposes. Discouraged use cases include real-time trading systems, automated investment decision-making, and high-stakes financial prediction.

2nd Supplementary Data
Usage for the dataset includes analyzing long-term iPhone sales and revenue trends, evaluating Apple's product and pricing strategy, and supporting educational, exploratory, and business intelligence analyses. Discouraged use cases include real-time sales forecasting, official financial reporting, and investment decision-making.

3nd Supplementary Data

Usage for the dataset includes analyzing global smartphone market trends, comparing iOS and Android market share, studying user adoption patterns, and conducting market competition research. Discouraged use cases include real-time market prediction, commercial market intelligence products, and high-stakes financial or strategic decision-making.

Licensing

Primary Data
CC BY-NC-SA 4.0
Data can be shared and adapted
Credit must be given to the author, only permitted for non-commercial use, modified dataset falls under the same license.

1st Supplementary Data
CC BY-NC-SA 4.0
The data may be shared and adapted for non-commercial purposes with proper attribution to the dataset author. Any modified or derived datasets must be distributed under the same license, and changes should be clearly documented.

2nd Supplementary Data
CC BY-NC-SA 4.0
The data may be shared and adapted for non-commercial purposes with proper attribution to the original author. Any modified versions must be distributed under the same license, and changes should be clearly indicated.

3nd Supplementary Data
CC BY-NC-SA 4.0
The data may be shared and transformed for non-commercial use with appropriate credit. Derivative works must carry the same license, and usage for commercial or proprietary applications is restricted.

Privacy considerations

Primary Data
Dataset only includes publicly available data since Apple is a publicly traded company which means all of their financials are publicly accessible. No PII is included.

1st Supplementary Data
The dataset consists of publicly available financial news articles and metadata related to Apple Inc. No personally identifiable information (PII) is included. All content is derived from public news sources and analyzed at the article level without reference to individual users.

2nd Supplementary Data
The dataset contains aggregated, product-level and company-level information about iPhone releases, estimated sales, and revenue. No personally identifiable information (PII) is included, and all figures are reported or estimated at an annual level.

3nd Supplementary Data

The dataset includes aggregated global statistics on smartphone users, sales, and market share. No individual-level data or personally identifiable information (PII) is present, and all values are reported in summarized form.

Format

The primary dataset is provided in a CSV file.
The first supplemental dataset is provided in a CSV file.
The second supplemental dataset is provided in a xlsx file.
The third supplemental dataset is provided in a CSV file.

Source and collection process

Origin

Primary Data
Yahoo Finance- https://finance.yahoo.com/quote/AAPL/history/
Nasdaq Apple Official Historical Data Table - https://www.nasdaq.com/market-activity/stocks/aapl/historical

- 1st Supplementary Data
Apple Stock (AAPL): Historical Financial News Data
Source: Kaggle
https://www.kaggle.com/datasets (Apple AAPL financial news dataset, 2016–2024)

- 2nd Supplementary Data
Apple iPhone Product & Revenue Dataset (2007–2025)
Source: Kaggle
https://www.kaggle.com/datasets/atharvamohite4631/apple-data-set-salesrevenuestock-priceprice

- 3nd Supplementary Data
Global iPhone & Smartphone Market Dataset (2011–2023)
Source: Kaggle
https://www.kaggle.com/datasets/mohamedfahim003/global-iphone-and-smartphone-market-2011-2023

Original datasets author/publisher

Primary Data
Yahoo Finance - Yahoo Inc. (Yahoo Finance Data Team)
Nasdaq Apple Official Historical Data Table - Nasdaq

- 1st Supplementary Data
Kaggle Community Dataset (published via Kaggle)

- 2nd Supplementary Data
Atharva Mohite (Kaggle dataset publisher)

- 3nd Supplementary Data
Mohamed Fahim (Kaggle dataset publisher)

Original datasets point of contact

Primary Data
Yahoo Finance
- Yahoo Finance Support Portal: https://help.yahoo.com
- General Yahoo Customer Care: https://help.yahoo.com/contact

Nasdaq
- Nasdaq Market Data Support: https://www.nasdaq.com/about/contact-us
- Market Data Inquiries Email: datasales@nasdaq.com

- 1st Supplementary Data
Dataset: Apple Stock (AAPL) Historical Financial News Data
Publisher: Frank Ossai (Kaggle)

- 2nd Supplementary Data
Dataset: Apple Data Set (Sales, Revenue, Stock Price)
Publisher: Atharva Mohite (Kaggle)

- 3nd Supplementary Data
Dataset: Global iPhone & Smartphone Market (2011–2023)
Publisher: Mohamed Fahim (Kaggle)

Original metadata

Primary Data
Apple Comprehensive Financial Dataset (1980–2026) — Metadata Page
https://www.kaggle.com/datasets/adamvakar/apple-comprehensive-financial-dataset-1980-2026

- 1st Supplementary Data
Apple Stock (AAPL) Historical Financial News Data
Metadata Link:
https://www.kaggle.com/datasets/frankossai/apple-stock-aapl-historical-financial-news-data

- 2nd Supplementary Data
Apple Data Set (Sales, Revenue, Stock Price)
Metadata Link:
https://www.kaggle.com/datasets/atharvamohite4631/apple-data-set-salesrevenuestock-priceprice

- 3nd Supplementary Data

Global iPhone & Smartphone Market (2011–2023)
Metadata Link:
https://www.kaggle.com/datasets/mohamedfahim003/global-iphone-and-smartphone-market-2011-2023

Collection process

- Primary Data
Yahoo
- Yahoo Finance - Price Data Adjustments & Methodology - https://help.yahoo.com/kb/SLN28256.html
- Yahoo Finance Historical Data (AAPL) - Source Table - https://finance.yahoo.com/quote/AAPL/history/

Nasdaq - Official Market Data Methodology
- Nasdaq Market Data Products - Collection & Processing Overview - https://www.nasdaq.com/solutions/market-data
- Nasdaq Data Collection & Dissemination - Technical Documentation - https://www.nasdaqtrader.com/Trader.aspx?id=MarketData
- Nasdaq AAPL Historical Data (Source Table) - https://www.nasdaq.com/market-activity/stocks/aapl/historical

- 1st Supplementary Data
https://www.kaggle.com/datasets/frankossai/apple-stock-aapl-historical-financial-news-data?utm_source=chatgpt.com
Collected and compiled by the dataset author using publicly available financial news articles related to Apple Inc., as described on the Kaggle dataset page.

- 2nd Supplementary Data
https://www.kaggle.com/datasets/atharvamohite4631/apple-data-set-salesrevenuestock-priceprice?resource=download
Collected and synthesized from publicly available Apple financial reports and market data, with post-2018 figures estimated by the author based on revenue and pricing trends, as documented on the Kaggle dataset page.

- 3nd Supplementary Data
https://www.kaggle.com/datasets/mohamedfahim003/global-iphone-and-smartphone-market-2011-2023?utm_source=chatgpt.com
Aggregated by the dataset author from publicly available global smartphone market reports and industry statistics, as described on the Kaggle dataset page.

Online link(s) to information on data source's collection and labeling processes.

Collection or access date

- Primary Data
Accessed: January 2026

- 1st Supplementary Data

Accessed: January 2026

- 2nd Supplementary Data
Accessed: January 2026

- 3nd Supplementary Data
Accessed: January 2026

Refresh date/schedule

- Primary Data
Last refresh Jan 3, 2026. Scheduled to refresh annually.

- 1st Supplementary Data
Not specified; dataset appears to be a fixed historical snapshot.

- 2nd Supplementary Data
Not specified; single published version with projected values.

- 3nd Supplementary Data
Not specified; dataset represents a fixed historical range.

Last refresh date and refresh cadence, or N/A if there is only one version.

Exploration Study

Questions driving the exploration study

1) Is daily news sentiment associated with Apple's daily stock return?
2) Do extreme sentiment days correspond to higher trading volume?
3) Do sentiment spikes precede larger price volatility?
4) How has Apple's stock price evolved over time?
5) Is iPhone revenue dependency associated with stock performance?
6) Does relative iOS market share (vs Android) relate to Apple's stock performance?

Descriptive statistics

Primary Dataset:

Variable	Count	Mean	Std	IQR	Min	Max
close_price	182	29.78	61.57	21.88	0.043634	271.859985
open_price	182	28.39	59.28	21.23	0.043634	272.260010
trading_volume	182	1.93e10	1.62e10	1.72e10	7.33e8	8.52e10
avg_daily_return	182	0.001106	0.003928	0.005085	−0.010246	0.015527
return_volatility	182	0.025439	0.010922	0.014315	0.009645	0.077053

Supplemental Dataset 1

Variable	Count	Mean	Std	IQR	Min	Max
sentiment_score	29,737	0.50	0.58	0.99	-1.00	1.00

Supplemental Dataset 2

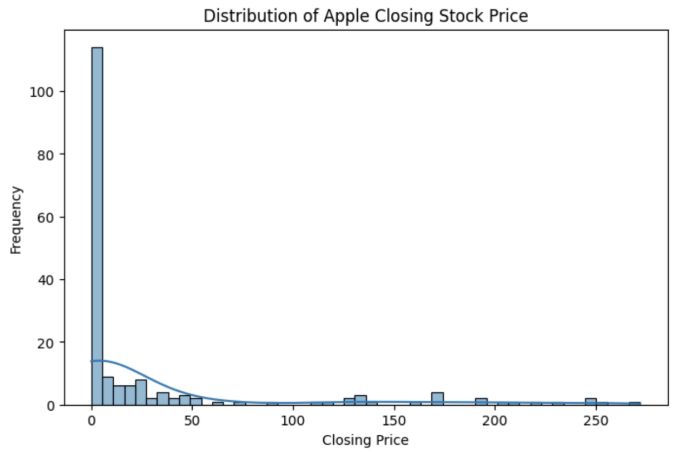
Variable	Count	Mean	Std	IQR	Min	Max
units_sold_in_year_millions	19	157.36	84.39	123.70	1.90	231.20
average_selling_price	19	719.95	157.17	225	450	945
annual_iphone_revenue_billions	19	124.63	75.99	120.58	0.95	218.30

Supplemental Dataset 3

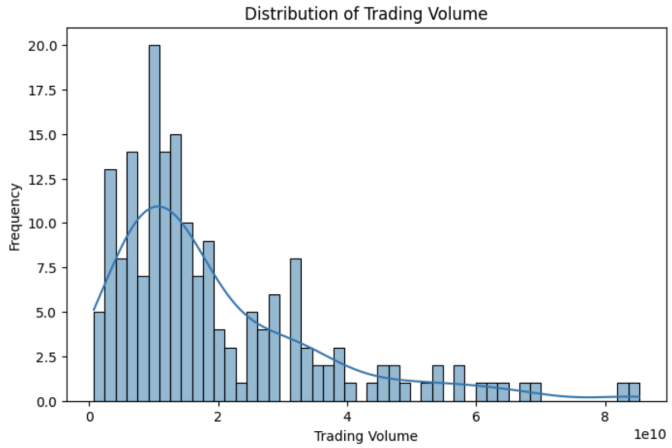
Variable	Count	Mean	Std	IQR	Min	Max
no_of_iphone_users	13	7.75e08	4.31e08	5.58e08	1.15e08	1.46e09
no_of_iphone_sold	13	1.95e08	4.31e07	3.50e07	9.31e07	2.36e08

ios_market_share	11	55.09	2.83	4.43	50.85	59.54
android_market_share	11	43.06	2.34	3.67	39.25	46.42

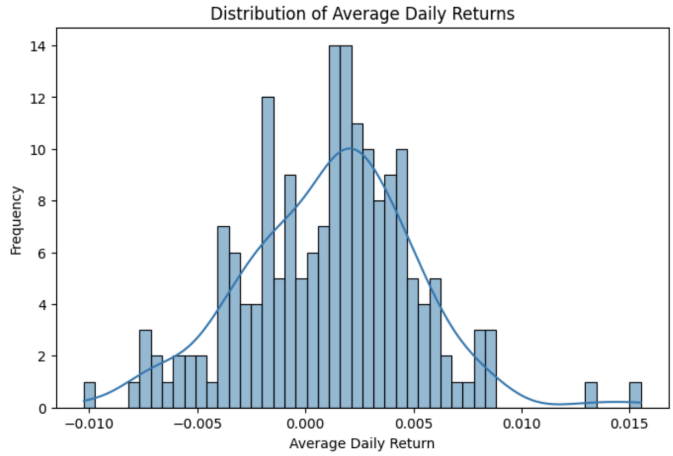
Distribution plots



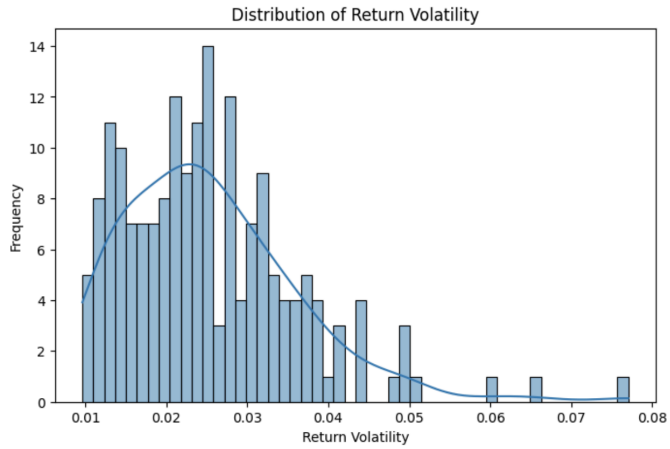
closing_price
The distribution of Apple's closing stock price is highly right-skewed, indicating that stock prices were relatively low during earlier years and increased significantly over time. This pattern reflects Apple's long-term financial growth and increasing market valuation.



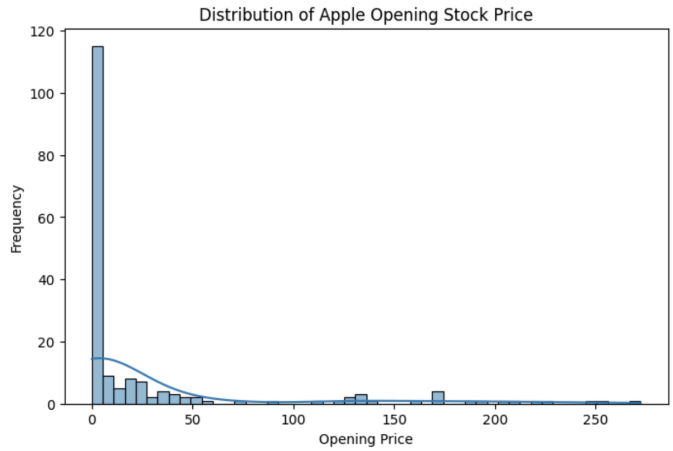
total_volume
Trading volume shows high variability with several extreme spikes. This suggests periods of increased investor activity, which may correspond to major company announcements, product launches, or broader market events. The data is skewed to the right, suggesting an overall lower trading volume apart from a few outliers.



avg_daily_return
The distribution of average daily returns is centered around zero, showing that gains and losses occur with similar frequency. Its relatively symmetrical bell shape reflects typical stock-market return behavior. Most values cluster close to zero, indicating low day-to-day volatility. The light tails show that extreme return swings are rare.

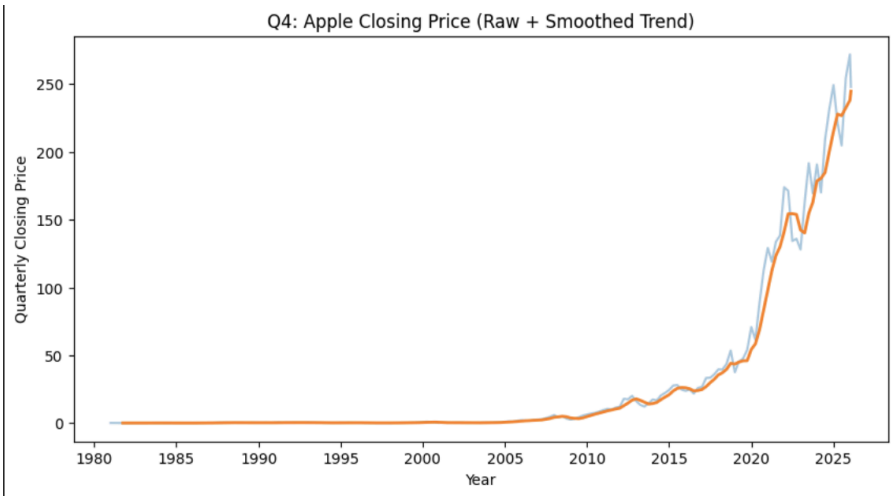


return_volatility
Return volatility is right-skewed, with most values clustering between 0.01 and 0.03, which reflects typical, low-to-moderate daily risk levels for a stable large-cap stock like Apple. A smaller number of periods show volatility rising above 0.04, indicating noticeably elevated uncertainty. The long right tail highlights rare spikes above 0.06, which signal unusually turbulent market conditions.

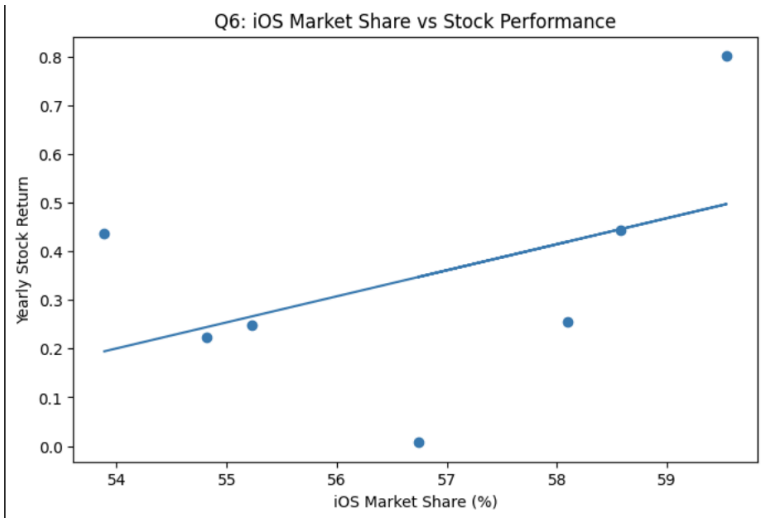


open_price
Apple's opening prices are heavily right-skewed, with most values clustered under $20 due to historical stock splits. The long tail toward higher prices reflects the company's strong growth over time. Only a few observations appear in the triple-digit range, showing that high opening prices are relatively recent. The distribution highlights Apple's shift from a low-priced stock to a much higher-valued one.

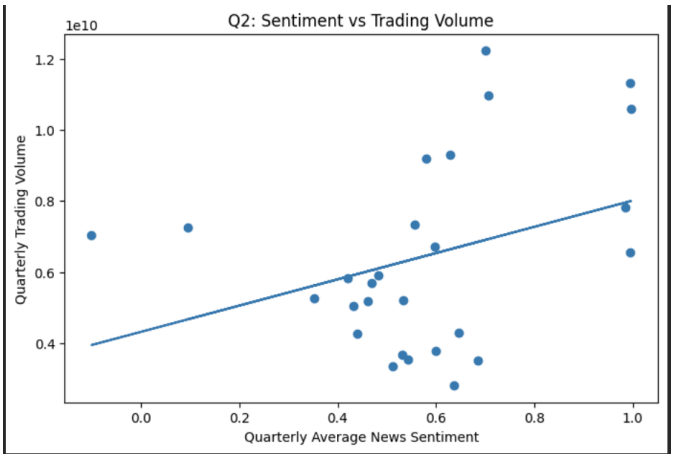
Visual analysis



Multiple time-series charts, especially the long-term stock price trend and iPhone revenue/units sold, show that Apple's financial growth is structurally linked to the iPhone product cycle. The stock price accelerates sharply after the iPhone's introduction, and yearly revenue visuals confirm the iPhone dominated Apple's earnings for more than a decade. These charts highlight Apple's transition from a diversified tech company to one whose valuation is strongly shaped by a single flagship product line.



The iOS vs. Android market share visualizations reveal a surprisingly stable and slightly rising iOS share over time, despite intense global competition. This stability, paired with consistent growth in global iPhone users, suggests Apple's ecosystem creates long-term customer lock-in. Across plots, the steadiness of iOS adoption aligns with Apple's sustained financial performance.



Across the scatterplots and time-series comparisons, sentiment showed a moderate positive relationship with trading volume, meaning quarters with unusually high or low sentiment tended to experience higher trading activity. This pattern suggests that sentiment doesn't directly influence returns, but it does increase investor attention and market participation. When sentiment spikes, whether strongly positive or strongly negative, investors react by trading more, resulting in noticeable volume increases during those periods.

Across scatterplots and normalized time series, a clear pattern emerges: extreme sentiment periods correlate with higher trading volume. Even though sentiment does not predict returns, the charts show that investor activity intensifies during highly positive or negative periods. This indicates sentiment acts as a market attention mechanism, affecting participation rather than price direction.

Stock return and volatility visualizations show that short-term returns behave randomly, with no clear autocorrelation patterns. However, the long-term stock price trend chart reveals smooth, compounding growth. This contrast highlights that while Apple's year-to-year performance is unpredictable, the long-term trajectory is strongly upward, which is consistent with its ecosystem strength, revenue base, and brand perception.

Analytical & statistical methods

To analyze temporal patterns in the Apple datasets, I applied two domain-specific time-series methods from the statsmodels.tsa library: Autocorrelation and Partial Autocorrelation (ACF/PACF) and the Augmented Dickey-Fuller (ADF) Unit Root Test. ACF and PACF were used to evaluate the structure of Apple's yearly stock returns, measuring whether returns depend on prior years or display any persistent autoregressive pattern. The results showed minimal autocorrelation and no meaningful PACF spikes, indicating that Apple's yearly returns behave like a random walk. This suggests that past performance does not reliably predict future outcomes and that Apple's long-run valuation is driven primarily by new market information rather than serial dependence.

The ADF test was applied to the quarterly average news sentiment series to determine whether sentiment is stationary or trending over time. The test produced a p-value greater than 0.05, meaning the series is non-stationary and exhibits a long-term upward drift rather than fluctuating around a constant mean. This indicates that Apple's media sentiment has gradually strengthened over time, reflecting increasing positive public perception and sustained brand momentum. Together, these two analytical methods provided statistical confirmation that Apple's financial performance shows little autoregressive structure, while sentiment shows persistent structural improvement.

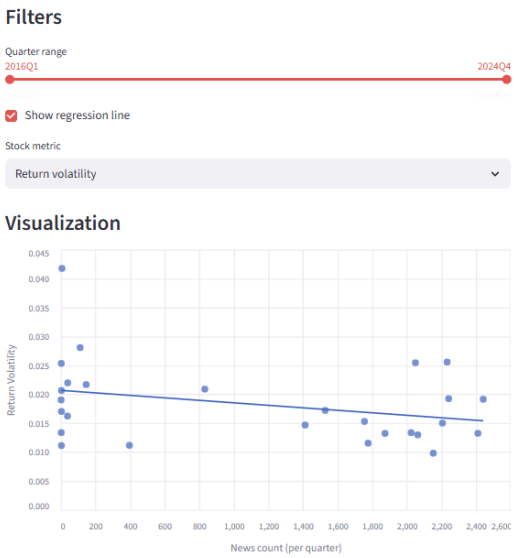
Polished visualizations

Q1 (Sentiment vs Return), I refined the visualization by applying several textbook design principles. First, I maximized the data-ink ratio by removing unnecessary legends, borders, and extra gridlines, ensuring that only meaningful visual elements remain. I used a uniform point style to reduce clutter and prevent unintended categorical interpretations. To guide attention, I used strategic color highlighting: muted blue points for the data and a darker, thicker line for the regression trend. I also relied on tooltips rather than on-chart text to keep the visualization clean while preserving analytical depth. These refinements make the

chart more readable, visually balanced, and aligned with best practices for analytical storytelling.

Q5 (sentiment vs stock performance), I refined the visualization into a clean scatterplot with an optional regression line. I maximized the data–ink ratio by using a single point style without legends or extra encodings, so each mark directly represents one quarter's sentiment and performance. An optional regression line highlights the overall trend without obscuring the raw data. I simplified axis titles and removed on-chart labels, relying instead on tooltips for details to keep the graphic uncluttered. These changes make the relationship between sentiment and stock performance easier to see and interpret, while still allowing deeper exploration when needed.

Q6, I refined the sentiment–performance visualization into a clean scatterplot with an optional regression line. I focused on maximizing the data–ink ratio by using a single point style and removing unnecessary legends or extra encodings, so each mark directly represents one quarter's sentiment and stock outcome. An optional regression line provides a clear summary of the overall relationship without obscuring the individual points. I simplified the axis titles and used tooltips instead of on-chart labels to reduce clutter while still allowing detailed inspection of each quarter. These choices make the relationship between sentiment and the selected stock metric (return or volatility) easier to interpret and more aligned with best-practice visual storytelling.



LLM-generated visualizations

1) ‘Does extreme sentiment correspond to higher trading volume?’

Feasibility Check

Feasible: `True`

Reason: The user is asking about the relationship between extreme sentiment and trading volume, which can be answered by the q2_extreme_sentiment_vs_volume dataset.

Code Validation

Generated code executed successfully: `True`

Description

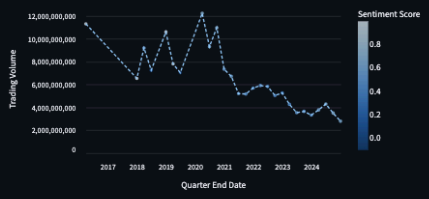
This chart visualizes the relationship between trading volume and sentiment score over time. A scatter plot shows the trading volume for each quarter, colored by the sentiment score. A dashed line overlay indicates the trend of trading volume across quarters. This helps to understand if extreme sentiment (both positive and negative) corresponds to higher trading volume.

Selected Dataset

q2_extreme_sentiment_vs_volume

Generated Chart

Trading Volume vs. Sentiment Score Over Time



Generated Code

```
import altair as alt

base = alt.Chart(df).encode(
    x=alt.X('quarter_end_date:T', axis=alt.Axis(title='Quarter End Date')),
    tooltip=['quarter_label', 'sentiment_score', 'trading_volume']
)

scatter = base.mark_circle().encode(
    y=alt.Y('trading_volume:Q', axis=alt.Axis(title='Trading Volume')),
    color=alt.Color('sentiment_score:Q', legend=alt.Legend(title='Sentiment

line = base.mark_line(strokeDash=[5,3]).encode(
    y=alt.Y('trading_volume:Q')
)

final_chart = (scatter + line).properties(title='Trading Volume vs. Sentime
```

Response time: 2.82 seconds

This visualization was generated by an LLM-based agent that first checked whether the request was feasible with the available data, selected the most appropriate dataset, generated Altair code, verified that the code executed successfully, and then rendered the chart in Streamlit.

2) ‘What is Apple’s trend in stock compared to sentiment over time?’

Feasibility Check

Feasible: `True`

Reason: The user is asking for a comparison of stock trends and sentiment over time, which can be addressed by plotting sentiment score against quarterly return over time. The `q1_sentiment_vs_return` dataset is suitable for this purpose as it contains both `sentiment_score` and `q_return` along with `quarter_end_date` for time-series analysis.

Code Validation

Generated code executed successfully: `True`

Description

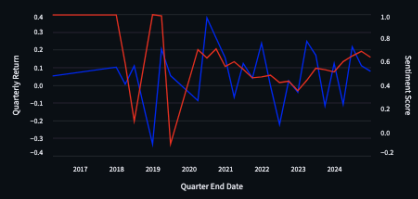
This chart visualizes Apple's quarterly stock return and average sentiment score over time. The x-axis represents the quarter end date, the left y-axis shows the quarterly stock return (in blue), and the right y-axis displays the average sentiment score (in red). Tooltips provide specific details for each data point when hovered over.

Selected Dataset

q1_sentiment_vs_return

Generated Chart

Apple's Quarterly Return vs. Sentiment Score Over Time



Generated Code

```
import altair as alt

base = alt.Chart(df).encode(
    x=alt.X('quarter_end_date:T', axis=alt.Axis(title='Quarter End Date')),
    tooltip=['quarter_label', 'q_return', 'sentiment_score']
)

line_return = base.mark_line(color='blue').encode(
    y=alt.Y('q_return', axis=alt.Axis(title='Quarterly Return'))
)

line_sentiment = base.mark_line(color='red').encode(
    y=alt.Y('sentiment_score', axis=alt.Axis(title='Sentiment Score'))
)

# Combine the two lines using a dual-axis chart
final_chart = alt.layer(line_return, line_sentiment).resolve_scale(y='indep
final_chart = final_chart.properties(title="Apple's Quarterly Return vs. Se
```

Response time: 3.04 seconds

This visualization was generated by an LLM-based agent that first checked whether the request was feasible with the available data, selected the most appropriate dataset, generated Altair code, verified that the code executed successfully, and then rendered the chart in Streamlit.

3) ‘Show the association between volatility and returns’

Feasibility Check

Feasible: `True`

Reason: The user is asking to see the association between volatility and returns, which can be answered by the 'q1_sentiment_vs_return' dataset as it contains both 'return_volatility' and 'q_return' columns.

Code Validation

Generated code executed successfully: `True`

Description

This scatter plot visualizes the association between return volatility and quarterly returns. Each point represents a quarter, with the x-axis showing the return volatility and the y-axis showing the quarterly return. Tooltips provide specific details for each quarter.

Selected Dataset

q1_sentiment_vs_return

Generated Chart

Generated Code

```
import altair as alt

base = alt.Chart(df).encode(
    x=alt.X('return_volatility', title='Return Volatility')
)

scatter = base.mark_circle().encode(
    y=alt.Y('q_return', title='Quarterly Return'),
    tooltip=['quarter_label', 'return_volatility', 'q_return']
)

final_chart = scatter
```

Response time: 2.49 seconds

This visualization was generated by an LLM-based agent that first checked whether the request was feasible with the available data, selected the most appropriate dataset, generated Altair code, verified that the code executed successfully, and then rendered the chart in Streamlit.

4) ‘Does news volume relate to Apple's overall financial success?’

Description

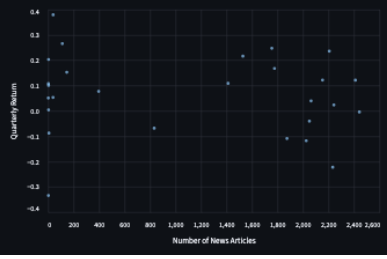
This visualization explores the relationship between news volume and Apple's financial performance. The top scatter plot shows the correlation between the number of news articles in a quarter and the quarterly stock return. The middle line chart tracks the average sentiment score over time, and the bottom line chart displays the trend of news volume over the same period. This allows for an analysis of whether increased news volume or specific sentiment trends are associated with changes in stock returns.

Dataset Used

q6_news_volume_vs_performance

Generated Chart

News Volume vs. Quarterly Return



Sentiment Score Over Time

News Volume Over Time

Generated Code

```
import altair as alt

# Assuming df is your pandas DataFrame with the specified schema
# df = pd.read_csv('your_data.csv') # Replace with your data loading

# Create the scatter plot
scatter = alt.Chart(df).mark_circle().encode(
    x=alt.X('news_count', title='Number of News Articles'),
    y=alt.Y('q_return', title='Quarterly Return'),
    tooltip=['quarter_label', 'news_count', 'q_return', 'sentiment_score']
).properties(
    title='News Volume vs. Quarterly Return'
)

# Create the line chart for sentiment score over time
line_sentiment = alt.Chart(df).mark_line(point=True).encode(
    x=alt.X('quarter_end_date', title='Quarter End Date'),
    y=alt.Y('sentiment_score', title='Average Sentiment Score'),
    tooltip=['quarter_label', 'sentiment_score']
).properties(
    title='Sentiment Score Over Time'
)

# Create the line chart for news count over time
line_news = alt.Chart(df).mark_line(point=True).encode(
    x=alt.X('quarter_end_date', title='Quarter End Date'),
    y=alt.Y('news_count', title='Number of News Articles'),
    tooltip=['quarter_label', 'news_count']
).properties(
    title='News Volume Over Time'
)

# Combine the charts vertically
final_chart = alt.vconcat(scatter, line_sentiment, line_news)
```

Response time: 3.7 seconds

This visualization was generated by an LLM-based agent that first checked whether the request was feasible with the available data, selected the most appropriate dataset, generated Altair code, verified that the code executed successfully, and then rendered the chart in Streamlit.

Cleaning and processing

Missing values

To identify missing values we used .isnull() and .sum() to determine the extent and location of incomplete data. In numerical columns, missing values were handled using certain strategies depending on the data context. For time-series variables, interpolation or forward/backgward filling was applied to maintain trends and continuity. For short gaps we used forward fill due to our assumption of short-term stability. In situations where missing observations were minimal or not critical, the rows were removed using .dropna(). We used these methods to ensure that all data points were filled and consistent for future analysis.

Aggregation

We used aggregation to summarize the dataset at meaningful levels for analysis. Using groupby() and resample() functions, data was aggregated across times (monthly/yearly totals) and across categorical variables (state or region). Common aggregation metrics included sums, means, and totals. sum() was used to aggregate sales-related data to reflect total monthly performance, while mean() was used to reflect averages of continuous environmental variables. This step helped reduce data granularity, which allowed for clear trend analysis/comparisons across groups while maintaining the integrity of KPIs.

Joins of input sources

The final step of preparing our data set included merging and joining operations in pandas across multiple datasets. The datasets were combined based on shared keys such as date or categorical identifiers. Appropriate join types like inner, left, or outer joins were selected depending on whether only matching records or all observations were required. After merging, the resulting dataset was validated to check for introduced missing values or inconsistencies. This integration helped further our analysis by combining complementary features from multiple data sources into a single analytical framework.